

PROTOPLANETARY DISKS

Protoplanetary disks are rotating structures of gas and dust surrounding newly formed stars. They are the natural outcome of star formation and represent the sites where planets are born. These disks typically survive for about 3 to 10 million years, during which dust grains grow, collide, form planetesimals, and eventually give rise to planets.

A typical protoplanetary disk consists of approximately 99% gas, primarily hydrogen and helium, and about 1% dust made of silicates, carbonaceous material, and various ices. The disk may extend from fractions of an astronomical unit (AU) close to the star to several hundred AU in the outer regions.

The disk is usually described in cylindrical coordinates (R, ϕ, z), where R is the radial distance from the star measured in the disk midplane, ϕ is the azimuthal angle, and z is the height above or below the midplane. Since protoplanetary disks are approximately axisymmetric, variations with respect to ϕ are often neglected, reducing the problem to a two-dimensional description involving only radius and height.

A protoplanetary disk is best modeled as a geometrically thin, hydrostatic, flared structure undergoing slow viscous evolution. Its radial structure is primarily controlled by the surface density and temperature distributions, while its vertical structure is governed by hydrostatic equilibrium and thermal pressure support.

Remarkably, a small set of equations describing Keplerian rotation, temperature, sound speed, scale height, and Gaussian vertical density is

sufficient to reproduce the essential morphology of observed protoplanetary disks. More sophisticated models such as DIANA, ProDiMo, and MCFOST build upon this foundation by incorporating dust physics, chemistry, radiative transfer, and planet-disk interactions.

THEORETICAL FRAMEWORK AND LITERATURE REVIEW

To develop a computational model of a protoplanetary disk, a comprehensive review of classical and modern studies describing disk structure, evolution, and planet-disk interactions was conducted.

VISCOUS EVOLUTION OF PROTOPLANETARY DISKS

The seminal work of Lynden-Bell and Pringle (1974) provides the fundamental framework for understanding the evolution of viscous circumstellar disks. Their study demonstrated that viscosity within the disk transports angular momentum outward while allowing mass to accrete inward toward the central star. As a consequence, the disk spreads radially over time.

The study assumes that the disk is geometrically thin, axisymmetric, and dominated by the gravitational potential of the central star. Under these assumptions, the angular velocity follows a Keplerian profile:

$$\Omega(R) = \sqrt{\frac{GM}{R^3}}$$

where M is the stellar mass and R is the cylindrical radius.

This paper established the basis for modern surface density prescriptions and viscous evolution equations. While the present work focuses on static disk structures rather than time-dependent evolution, the concepts introduced in this study motivate the adopted radial density distributions.

PASSIVE IRRADIATED FLARED DISKS

Chiang and Goldreich (1997) proposed a hydrostatic model for passive circumstellar disks surrounding T-Tauri stars. Their principal contribution was demonstrating that protoplanetary disks naturally assume a flared geometry when heated by stellar irradiation.

The disk is represented by two distinct layers:

- * An optically thin superheated surface layer directly exposed to stellar radiation.
- * A cooler, optically thick interior layer heated by reprocessed radiation from the surface.

The temperature distribution approximately follows a power law:

$$T(R) = T_0 \left(\frac{R}{1 \text{ AU}} \right)^{-q}$$

where T_0 is a reference temperature and the power-law index q typically ranges between 0.4 and 0.6.

Assuming vertical hydrostatic equilibrium, the local pressure scale height is expressed as:

$$H(R) = \frac{c_s}{\Omega}$$

where the isothermal sound speed c_s is defined as:

$$c_s = \sqrt{\frac{k_B T}{\mu m_H}}$$

Here, k_B is the Boltzmann constant, μ is the mean molecular weight, and m_H is the mass of a hydrogen atom. This formulation naturally leads to a flared disk geometry in which the aspect ratio H/R increases with distance from the star.

VERTICAL DENSITY STRUCTURE

Under the assumption of hydrostatic equilibrium, vertical pressure gradients balance the vertical component of stellar gravity. Assuming an isothermal vertical profile, the gas density follows a Gaussian distribution:

$$\rho(R, z) = \frac{\Sigma(R)}{\sqrt{2\pi}H(R)} \exp\left(-\frac{z^2}{2H(R)^2}\right)$$

where:

- * $\Sigma(R)$ is the surface density,
- * $H(R)$ is the pressure scale height,
- * z is the vertical coordinate measured from the midplane.

This equation forms the basis for constructing two-dimensional density maps and visualizing the vertical morphology of the disk.

VISCOSITY PARAMETERIZATION

Although the physical origin of angular momentum transport remains an active area of research (e.g., magneto-rotational instability), most disk models adopt the Shakura-Sunyaev viscosity prescription:

$$\nu = \alpha c_s H$$

where α is a dimensionless turbulence parameter. Typical values for protoplanetary disks lie in the range:

$$10^{-4} \leq \alpha \leq 10^{-2}$$

In the present study, viscosity is treated primarily as a parameter governing disk structure and potential future extensions involving viscous evolution, rather than a dynamically evolving quantity.

PLANET-DISK INTERACTIONS

Duffell (2015) introduced an analytical model describing gaps carved by planets embedded within protoplanetary disks. Gap formation results from the competition between planetary gravitational torques and the viscous redistribution of gas.

The depth of a gap can be approximated by the ratio of the surface density at the planet's location (Σ_p) to the unperturbed surface density (Σ_0):

$$\frac{\Sigma_p}{\Sigma_0} = \frac{1}{1 + 0.04K}$$

where the dimensionless parameter K is defined as:

$$K = \frac{q^2}{\alpha} \left(\frac{R}{H} \right)^5$$

and:

- * $q = M_p/M_*$ is the planet-to-star mass ratio,
- * α is the viscosity parameter.

Although gap formation is beyond the scope of the baseline model developed here, these relations provide a robust framework for the future implementation of planet-induced structures.

MODELING PHILOSOPHY

Mordasini et al. (2014) presented a global framework for planet formation and evolution. Rather than introducing new disk equations, this work emphasizes a modular approach in which multiple physical processes are combined into a unified simulation pipeline.

Inspired by this methodology, the current project adopts a modular architecture consisting of independent components responsible for:

- * Surface density calculations
- * Temperature profile generation
- * Scale height determination
- * Vertical density reconstruction
- * Rotational velocity estimation
- * Gap generation (optional)
- * Visualization utilities

This structure allows for the straightforward incorporation of additional physics, such as dust evolution, radiative transfer, chemistry, and population synthesis, in future developments.

SUMMARY OF ADOPTED PHYSICAL MODEL

The present work utilizes a simplified yet physically motivated description of a protoplanetary disk based on five key equations:

1. Keplerian angular velocity:

$$\Omega = \sqrt{\frac{GM_*}{R^3}}$$

2. Radial temperature profile:

$$T(R) = T_0 \left(\frac{R}{1 \text{ AU}} \right)^{-q}$$

3. Isothermal sound speed:

$$c_s = \sqrt{\frac{k_B T}{\mu m_H}}$$

4. Pressure scale height:

$$H = \frac{c_s}{\Omega}$$

5. Vertical Gaussian density profile:

$$\rho(R, z) = \frac{\Sigma(R)}{\sqrt{2\pi}H} e^{-z^2/(2H^2)}$$

Together, these equations provide a computationally efficient representation of a flared, hydrostatic, passively heated protoplanetary disk and form the theoretical basis of the numerical simulations developed in this project.

ASSUMPTIONS AND SIMPLIFICATIONS ADOPTED IN THE PRESENT MODEL

The objective of the present work is to develop a physically motivated static model of a protoplanetary disk that captures its fundamental radial and vertical structure while maintaining computational simplicity.

In order to achieve this, several assumptions and simplifications commonly employed in protoplanetary disk modelling are adopted.

GEOMETRICALLY THIN DISK APPROXIMATION

The disk is assumed to be geometrically thin, implying that its vertical extent is significantly smaller than its radial extent, i.e.,

$$H \ll R$$

where H is the pressure scale height and R is the cylindrical radius.

This assumption allows the radial and vertical structures of the disk to be treated independently to first order and is well justified for most observed protoplanetary disks.

AXISYMMETRIC DISK STRUCTURE

The disk is assumed to possess rotational symmetry about the central star. Consequently, all physical quantities are considered independent of the azimuthal coordinate ϕ .

Only radial and vertical variations are considered,

$$\rho = \rho(R, z)$$

This assumption neglects spiral arms, vortices, asymmetries, warps, and local perturbations that may arise due to embedded planets or instabilities.

STATIC DISK APPROXIMATION

The present model describes a static snapshot of a disk rather than its temporal evolution.

The quantities considered in the model are therefore independent of time,

$$\partial/\partial t = 0$$

Processes such as viscous spreading, accretion evolution, photoevaporation, and disk dispersal are neglected for now.

This approximation is sufficient for investigating the overall density and thermal structure of a typical protoplanetary disk.

KEPLERIAN ROTATION

It is assumed that the stellar mass dominates the gravitational potential,

$$M(\text{disk}) \ll M(\text{star})$$

The gas therefore follows Keplerian rotation,

$$\Omega = \sqrt{GM(\text{star})/R^3}$$

Self-gravity of the disk is neglected.

This assumption is valid for the majority of T-Tauri and Herbig disks whose masses are generally much smaller than the stellar mass.

VERTICAL HYDROSTATIC EQUILIBRIUM

The disk is assumed to be in hydrostatic equilibrium in the vertical direction.

At every radius, the vertical pressure gradient balances the vertical component of stellar gravity.

This assumption leads naturally to the determination of the pressure scale height and a Gaussian vertical density profile.

Rapid vertical motions and turbulence-driven deviations from equilibrium are neglected.

VERTICALLY ISOTHERMAL APPROXIMATION

At a given radius, the temperature is assumed to remain constant with height.

Consequently, the vertical density distribution takes a Gaussian form.

In reality, disk atmospheres are often significantly hotter than the midplane due to direct stellar irradiation. However, assuming vertical isothermality considerably simplifies the construction of a first-order density model.

PASSIVE IRRADIATION DOMINATED DISK

Heating of the disk is assumed to arise primarily from stellar irradiation.

Viscous heating associated with accretion processes is neglected.

This approximation is appropriate for evolved T-Tauri disks exhibiting relatively low accretion rates and is widely used in passive flared disk models.

The temperature profile is prescribed using a simple radial power law.

GAS AND DUST ARE WELL MIXED

The model assumes that gas and dust remain perfectly coupled throughout the disk.

A constant dust-to-gas mass ratio of approximately 1:100 is adopted.

Processes including

- dust settling,
- radial drift,
- grain growth,
- fragmentation,

are neglected for now.

These effects may significantly alter disk structure but require substantially more sophisticated treatments.

CONSTANT MEAN MOLECULAR WEIGHT

The gas composition is assumed to remain spatially uniform.

A constant mean molecular weight

$$\mu = 2.3$$

is adopted, corresponding to a mixture primarily composed of molecular hydrogen and helium.

Chemical evolution and compositional gradients are not considered.

NEGLECT OF RADIATIVE TRANSFER AND CHEMISTRY

Detailed radiative transfer calculations are beyond the scope of the present model.

Consequently, the following processes are omitted:

- molecular chemistry,
- photodissociation,
- ionization,
- freeze-out,
- non-LTE effects,
- gas heating and cooling balance,
- line radiative transfer.

Instead, a prescribed temperature profile is utilized.

NEGLECT OF PLANET-DISK INTERACTIONS

The baseline model assumes a smooth disk without embedded planets.

Features such as

- gaps,
- rings,
- spiral density waves,
- cavities,

are not included.

However, these effects may be incorporated in future extensions through analytical prescriptions.

SURFACE DENSITY DISTRIBUTION

The radial distribution of disk material is described through the surface density $\Sigma(R)$, representing the vertically integrated mass per unit area.

Rather than solving the complete viscous evolution equation, an analytical self-similar profile motivated by viscous disk theory is adopted

$$\Sigma(R) = \Sigma_c (R/R_c)^{(-\gamma)} \exp[-(R/R_c)^{(2-\gamma)}]$$

Where :

Σ_c is a normalization constant,

R_c is the characteristic radius,

γ controls the radial slope.

This profile reproduces many observed disk structures and is widely used in modern disk modelling codes. It naturally exhibits a power-law behavior in the inner disk and an exponential taper at large radii.

HYDROSTATIC EQUILIBRIUM

The vertical structure of the disk is obtained by assuming hydrostatic equilibrium.

At every radius, the vertical pressure gradient balances the vertical component of stellar gravity.

This balance is described by

$$(1/\rho)(dP/dz) = -\Omega^2 z$$

This equation represents one of the most fundamental relations in disk theory.

Physically, gravity attempts to compress the disk toward the midplane, while gas pressure tends to expand it vertically.

The competition between these effects determines the disk thickness

SUMMARY OF ADOPTED ASSUMPTIONS

Assumption	Purpose
Thin disk	Simplifies geometry
Axisymmetry	Reduces problem to 2D
Static disk	Avoids time evolution
Keplerian rotation	Neglects disk self-gravity
Hydrostatic equilibrium	Determines vertical structure
Vertically isothermal	Allows Gaussian density profile
Passive heating	Simplifies temperature calculation
Analytical $\Sigma(R)$	Avoids solving viscous equations
Well-mixed gas and dust	Simplifies composition
Constant μ	Simplifies sound speed calculations
α -viscosity	Parameterizes turbulence
No chemistry	Reduces computational cost
No planets	Provides smooth reference model

These assumptions provide a balance between physical realism and computational simplicity, enabling the construction of a self-consistent first-order protoplanetary disk model while retaining the possibility of incorporating more advanced physics in future work.